Prehistory: the p-adic setting

26/02/26

Today: * quick recap on p-adic integrals
* Igusa's zeta function is rational
* Denef's formula

§1 Recap about p-adics

Why p-adics? • Example of integer solutions to $f(\underline{x}) = 0$ $f \in \mathbb{Z}[x_1, \dots, x_n]$

• If there is a (non-trivial) solution in $\mathbb{Z}$, then there is mod p^m for all powers of primes.

• $\mathbb{Z}/p^{m'}\mathbb{Z} \xrightarrow{\pi_{m',m}} \mathbb{Z}/p^m\mathbb{Z}$

Definition: * $\mathbb{Z}_p := \varprojlim_m \mathbb{Z}/p^m\mathbb{Z}$ $\qquad \mathbb{Q}_p = \mathrm{Frac}(\mathbb{Z}_p)$

* Equivalently, $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ w.r.t p-adic abs. value

Think of elements of $\mathbb{Q}_p$ as convergent power series in powers of p.
Partial sums are "Taylor expansions" of functions with local parameter p.

in $\mathbb{Q}_p$, $\frac{1}{1-p} = 1 + p + p^2 + \dots$

Properties: * $\mathbb{Q}_p$ loc. compact but totally disconnected top. space
* $\mathbb{Z}_p = \{|x| \leq 1\}$ open and closed, and compact
* $\mathbb{Z}_p$ local ring $\quad m_p = p\mathbb{Z}_p \quad \mathbb{Z}_p/m_p \simeq \mathbb{F}_p$
* basis of open and closed neighborhoods

$$a + p^m\mathbb{Z}_p \qquad a \in \mathbb{Q}_p \qquad m \in \mathbb{N}$$

* $\mathbb{Z}_p = \bigsqcup_{a \in \{0, \dots, p-1\}} (a + p\mathbb{Z}_p)$

§2 Affine p-adic integration in a nutshell

$\mathbb{Q}_p$ loc. comp. Ab. group $\rightsquigarrow$ admits a unique HAAR measure (up to scalar)

Definition. $\mu(a+p^m\mathbb{Z}_p) = \frac{1}{p^m}$ for any $a \in \mathbb{Q}_p$ and $m \in \mathbb{N}$.

$(\mu(\mathbb{Z}_p) = 1)$

Subsets of importance are <u>cylinders</u> : $C \subseteq \mathbb{Z}_p^d$ s.t. $C = \pi_m^{-1}(\pi_m(C))$ for some $m \geq 0$

(in geom. motivic integ: $C \subseteq \mathbb{C}[[t]]$ st. $C = \pi_m^{-1}(\pi_m(C))$)

$\pi_m : \mathbb{C}[[t]] \longrightarrow \mathbb{C}[t]/t^{m+1}$

Example: $p^{m+1}\mathbb{Z}_p^d = \pi_m^{-1}(0) = \pi_m^{-1}(\pi_m(p^{m+1}\mathbb{Z}_p))$

Fact: For any cylinder C, $\mu(C) = \frac{\#\pi_m(C)}{p^{d(m+1)}}$ for $m \gg 0$.

because $C = \bigsqcup_{a \in \pi_m(C)} a + (p^{m+1}\mathbb{Z}_p)^d$ for $m \gg 0$

An integral

$$\int_{\mathbb{Z}_p} |x^N|_p^s \, d\mu = \frac{p-1}{p-p^{-Ns}} \qquad N \in \mathbb{N} \qquad s \in \mathbb{C},\ \mathrm{Re}(s) > 0$$

$$\int_{\mathbb{Z}_p^d} |x_1^{N_1} \cdots x_d^{N_d}|^s \, d\mu = \prod_{i=1}^{d} \frac{p-1}{p-p^{-N_i s}}$$

$$\int_{\mathbb{Z}_p} |x^N|_p^s \, d\mu = \sum_{m \geq 0} \int_{p^m\mathbb{Z}_p \setminus p^{m+1}\mathbb{Z}_p} |x^N|_p^s \, d\mu = \sum_{m \geq 0} (p^{-m} - p^{-(m+1)}) p^{-mNs}$$

$$= \frac{p-1}{p-p^{-Ns}}.$$

Working locally and patching, one can define a theory of integration in $\mathbb{Q}_p$-analytic manifolds with measures attached to top deg diff forms

§ 3 Igusa's zeta function

Definition. $f \in \mathbb{Z}_p[X_1, \dots, X_d]$ $\quad s \in \mathbb{C}$ (formal variable)

$$\boxed{Z_f(s) = \int_{\mathbb{Z}_p^d} |f(x)|_p^s \, d\mu(x)}$$

(holo on $\mathrm{Re}(s) > 0$)

Theorem (IGUSA) $Z_f(s)$ is a rational function in q^{-s}.

Toolbox.

Theorem (embedded res. of singularities) $f \in \mathbb{Q}_p[X_1, \dots, X_d]$ non-constant

There exists $\pi : Y \longrightarrow \mathbb{A}^d_{\mathbb{Q}_p}$ $\quad \dim Y = d$

(1) $\mathrm{Exc}(\pi)$ SNC divisor (locally $\times$)

(2) $\mathrm{div}(\pi^* f) = \sum_i N_i E_i$

(3) $\mathrm{div}(\mathrm{Jac}(\pi)) = \mathrm{div}(\pi^*(dx_1 \wedge \dots \wedge dx_d)) = \sum_i (\nu_i - 1) E_i$

"numerical data"

$\nu_i \in \mathbb{Z}_{>0}$

obtained by a composition of blow-ups over smooth centers.

Theorem (change of variables) $\varphi : Y \to X$ $\mathbb{Q}_p$-analytic between compact

bianalytic outside a measure-0 closed subset Z

ω top-degree $\mathbb{Q}_p$-analytic diff form on X

$f : X \longrightarrow \mathbb{Q}_p$ $\mathbb{Q}_p$-analytic

$$\int_X |f|_p^s \, d\mu_\omega = \int_Y |f \circ \varphi|_p^s \, d\mu_{\varphi^*\omega}$$

$\|$ by def

$$p^{-s \, \mathrm{ord}_p(f(x))}$$

$\rightsquigarrow T^{\mathrm{ord}_t f(x)}$

Motivic setting:

$T = \mathbb{L}^{-s}$ later

Theorem (DENEF's formula) With the notations from before, assume that π has good reduction mod p. Then:

$$Z_f(s) = p^{-d} \sum_{I \subseteq \{0,\dots,r\}} \# E_I^\circ(\mathbb{F}_p) \prod_{i \in I} \frac{(p-1)\, p^{-N_i s - \nu_i}}{1 - p^{-N_i s - \nu_i}}$$

where $E_I^\circ = \bigcap_{i \in I} E_i \setminus \bigcup_{j \notin I} E_j$ for all I.

Proof: "local to global" We can write:

$$Z_f(s) = \sum_{\substack{j \\ (\text{finite})}} \int_{U_j} |y_1|^{N_{i_1} s + \nu_{i_1} - 1} \cdots |y_k|_p^{N_{i_k} s + \nu_{i_k} - 1} |u(s)|_p^s |v(y)|_p \, d\mu_{\pi^* dx_1 \wedge \dots \wedge dx_d}$$

* $(N_{i_1}, \nu_{i_1}), \dots, (N_{i_k}, \nu_{i_k})$ subset of num. data of $Y \to \mathbb{A}^d$
* $\pi^* f = y_1^{N_1} \cdots y_k^{N_{i_k}} u(y)$ $\quad u \in \mathcal{O}_{U_j}^\times$ $\quad 1 \leq k \leq d$
* $\pi^*(dx_1 \wedge \dots \wedge dx_d) = y_1^{\nu_{i_1} - 1} \cdots y_k^{\nu_{i_k} - 1} v(y)\, dy_1 \wedge \dots \wedge dy_d$, $v \in \mathcal{O}_{U_j}^\times$

One can be more precise by introducing

$$a \in Y(\mathbb{F}_p) \quad B_a = \{ y \in Y(\mathbb{Z}_p) \mid y = a \bmod \mathbb{F}_p \}$$

$$I_a = \{ i \mid a \in E_i(\mathbb{F}_p) \}$$

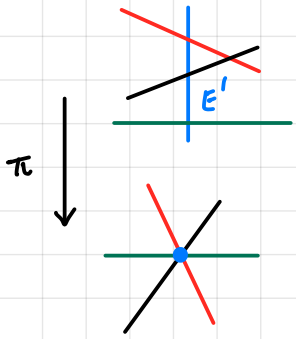
$\boxed{\text{assume } u, v \text{ are units}}$

Then $Z_f(s) = \sum_{a \in Y(\mathbb{F}_p)} \int_{B_a} \cdots = \sum_{a \in Y(\mathbb{F}_p)} p^{-d} \prod_{i \in I_a} \frac{(p-1)\, p^{-N_i s - \nu_i}}{1 - p^{-N_i s - \nu_i}}$.

$B_a \to p\mathbb{Z}_p^d$ biianalytic
$x \mapsto y_1(x) \dots y_d(x)$

$\square$

An example. $f(x,y) = xy(x+y)$



$E = E_1 + E_2 + E_3 + 3E'$

$N_\bullet = (1,1,1,3)$

$\nu_\bullet = (1,1,1,2)$

By hand one can show

$$Z_p(s) = \frac{(1-p^{-1})(1-2p^{-1})}{1-p^{-(3s+2)}} + 3p^{-s-1}\,\frac{1-p^{-1}}{1-p^{-s-1}} \cdot \frac{1-p^{-1}}{1-p^{-(3s+2)}}$$